

A Low Memory Footprint Quantized Neural Network for Depth Completion of Very Sparse Time-of-Flight Depth Maps

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Abstract

Sparse active illumination enables precise time-of-flight depth sensing as it maximizes signal-to-noise ratio for low power budgets. However, depth completion is required to produce dense depth maps for 3D perception. We address this task with realistic illumination and sensor resolution constraints by simulating ToF datasets for indoor 3D perception with challenging sparsity levels. We propose a quantized convolutional encoder-decoder network for this task. Our model achieves optimal depth map quality by means of input pre-processing and carefully tuned training with a geometry-preserving loss function. We also achieve low memory footprint for weights and activations by means of mixed precision quantization-at-training techniques. The resulting quantized models are comparable to the state of the art in terms of quality, but they require very low GPU times and achieve up to 14-fold memory size reduction for the weights w.r.t. their floating point counterpart with minimal impact on quality metrics.

1. Introduction

Time-of-flight (ToF) sensors are active depth sensing devices [53] with the potential to provide more reliable scene understanding by true 3D perception. Due to their low power consumption and accuracy at real-time frame rates, ToF sensors were recently integrated in mobile consumer devices [33]. However, ToF relies on active illumination [53] which accounts for a relevant part of its power consumption. To use the limited power budget of a mobile device more efficiently, the scene can be illuminated with a dot pattern light source so that its radiant intensity concentrates onto a small number of regions (*dots*). A low-power ToF sensor for indoors 3D perception typically captures 500 ~ 1500 dots per frame. Because of this sparsity level, sensor fusion techniques are necessary to obtain dense

depth maps. Hereafter, we tackle this *depth completion* task by using a single RGB image frame together with a sparse depth map warped to the RGB camera view (at small baseline of a few *mm* to minimize occlusion issues w.r.t. the ToF sensor). We dub this the *sparse ToF* setting.

To obtain dense depth from a sparse depth map, deep learning-based approaches have gained significant attention, with several contributions showing competitive performances especially on autonomous driving datasets [17, 44, 47]. We instead focus on efficient indoor 3D perception with ToF. In order to attain overall low power budget for both ToF sensing and depth completion, neural networks must be designed and trained accordingly for efficient inference. In this paper we introduce a carefully designed encoder-decoder CNN for depth completion. We achieve a low memory footprint for weights and activations by means of mixed-precision quantization-aware training and show that the resulting quantized models achieve conspicuous memory size savings for weights and activations, while remaining competitive w.r.t. quality metrics. Our contributions are as follows:

- We propose an efficient depth completion model suitable for low-power sparse ToF sensing. We evaluate our model and other state-of-the-art models on two datasets: the public-domain dataset NYU-Depth v2 [43], and SDS-ST1k [6], a contributed dataset providing accurate sparse ToF and RGB camera simulations.
- We analyze the impact of input pre-processing providing fast initialization; a normals estimation block enabling supervision with ground truth normals; different loss functions and network configurations. Each of these elements contributes to achieving optimal depth map quality for the resulting model.
- We evaluate quantization-aware training strategies to quantize weights and activations of the previous model. To the best of our knowledge, mixed-precision quantization [46] has been used successfully to classification tasks. In this paper we present the first ap-

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